

Article

Efficient Color Image Encoding Using Chroma-Subsampled Hybrid Quantum Representation (CS-HQR)

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Abstract

Quantum image representation (QIR) is fundamental for leveraging quantum computational advantages in image processing. However, existing encoding schemes face challenges including excessive qubit requirements, high gate complexity, and substantial circuit depth that render them impractical for near-term intermediate-scale quantum (NISQ) devices. Current methods treat all image channels equally, disregarding perceptual characteristics that classical compression standards have exploited for decades. This paper introduces CS-HQR (Chroma Subsampling Hybrid Quantum Representation), a novel quantum image encoding methodology bridging classical image compression with quantum circuit optimization. CS-HQR incorporates YCbCr color space transformation and 4:2:0 chroma subsampling, exploiting human visual perception's higher sensitivity to luminance over chrominance information. The dual-circuit architecture encodes luminance at full resolution using 16 qubits while chrominance channels use 15 qubits at quarter resolution, achieving 50% reduction in encoding complexity while maintaining SSIM above 0.98. Comprehensive experimental validation using 6,097 medical brain scan slices demonstrates CS-HQR achieves approximately 51% gate count reduction compared to MCQI, substantial circuit depth reduction, and 75% chrominance data reduction. Perceptual fidelity is validated via SSIM analysis on a representative 404-image subset, consistently exceeding 0.9999. The method exhibits superior scalability with $O(n \cdot 2^n)$ complexity versus $O(2^{2n})$ for amplitude-based methods.

Keywords: quantum image representation; chroma subsampling; NISQ devices; quantum circuits; medical image processing; perceptual image quality



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1. Introduction

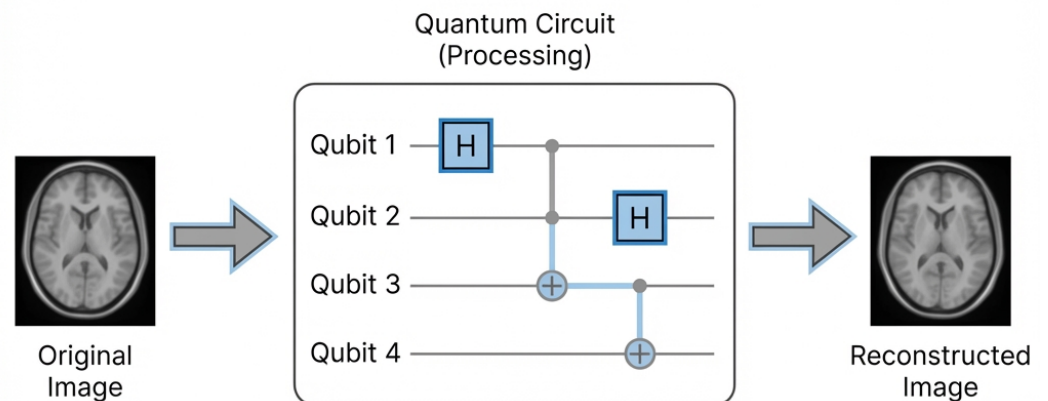


Figure 1. Overview of quantum image processing: bridging classical image data with quantum computational paradigms for enhanced processing capabilities.

The intersection of quantum computing and image processing represents one of the most promising frontiers in computational science. As medical imaging continues to generate unprecedented volumes of data with global healthcare imaging data projected to exceed 2,314 exabytes by 2026 [1] the computational demands for processing, analyzing, and transmitting these images strain classical computing infrastructure. Quantum computing offers a paradigm shift through its inherent parallelism, where n qubits can simultaneously represent 2^n states, potentially enabling exponential speedups for specific image processing tasks [2].

However, realizing this quantum advantage requires efficient methods to encode classical image data into quantum states a challenge that current quantum image representation (QIR) methods have yet to adequately address. Existing approaches such as FRQI [3] and NEQR [4] were designed with theoretical elegance prioritized over practical implementation constraints. FRQI, while qubit-efficient with only $n + 1$ qubits for 2^n pixels, requires circuit depths proportional to 2^{2^n} gates, rendering it impractical for images larger than 8×8 pixels on current hardware [5]. NEQR improves upon this with linear gate complexity but demands additional qubits for color depth encoding.

The critical observation motivating our work is that these methods uniformly ignore decades of research in perceptual image processing. Classical image and video compression standards JPEG, MPEG-2, H.264, HEVC achieve remarkable compression ratios by exploiting the differential sensitivity of human vision to luminance versus chrominance information [6]. The human visual system contains approximately 120 million rod cells sensitive to brightness but only 6 million cone cells for color perception, creating an inherent asymmetry that compression algorithms leverage through chroma subsampling [7].

This biological insight has never been systematically applied to quantum image representation. Current multi-channel methods like MCQI [8] treat red, green, and blue channels identically, encoding each at full resolution despite the perceptual redundancy this introduces.

Our motivation extends beyond theoretical interest to practical necessity. The current era of noisy intermediate-scale quantum (NISQ) devices imposes severe constraints: IBM's largest publicly accessible quantum processors offer approximately 127 qubits with coherence times measured in microseconds and two-qubit gate error rates around 1% [9]. Every unnecessary gate increases the probability of decoherence-induced errors, making circuit optimization essential for meaningful quantum computation.

Medical imaging presents particularly compelling use cases for quantum image processing. Diagnostic accuracy in radiology depends on preserving clinically relevant features while managing the computational burden of increasingly high-resolution modalities [10]. Brain MRI analysis, tumor detection, and radiomics applications demand processing pipelines that maintain diagnostic fidelity.

CS-HQR emerges from the recognition that quantum algorithm design should incorporate domain-specific knowledge rather than treating all data uniformly. By bridging established principles from classical signal processing with quantum circuit optimization, we demonstrate that significant practical improvements are achievable without sacrificing the theoretical foundations of quantum image representation.

1.1. Evolution of Quantum Image Representation

The Flexible Representation of Quantum Images (FRQI), proposed by Le et al. in 2011, represented a significant conceptual advance [3]. FRQI encodes grayscale pixel intensities as rotation angles applied to a single “color qubit.” The FRQI quantum state is expressed as:

$$|I\rangle_{\text{FRQI}} = \frac{1}{2^n} \sum_{i=0}^{2^{2n}-1} (\cos \theta_i |0\rangle + \sin \theta_i |1\rangle) \otimes |i\rangle \quad (1)$$

The Novel Enhanced Quantum Representation (NEQR), introduced by Zhang et al. [4], addressed FRQI’s probabilistic retrieval limitation:

$$|I\rangle_{\text{NEQR}} = \frac{1}{2^n} \sum_{y=0}^{2^n-1} \sum_{x=0}^{2^n-1} |C_{yx}\rangle \otimes |y\rangle \otimes |x\rangle \quad (2)$$

Subsequent developments introduced GQIR [?], MCQI [8], and QLR [?].

1.2. Limitations of Existing Approaches

Despite two decades of research, existing QIR methods share a common limitation: they treat all image information uniformly without considering perceptual relevance. Table 1 summarizes the key limitations.

Table 1. Limitations of Existing QIR Methods.

Method	Primary Limitations	Type
FRQI	Exponential gate complexity $O(2^{2n})$; impractical for images $> 8 \times 8$	Amplitude
NEQR	Linear but still substantial gate requirements; no compression	Basis State
GQIR	Arbitrary bit-depth but uniform treatment of all pixels	Basis State
MCQI	Tripled resource requirements for color; no perceptual optimization	Basis State
EFRQI	Improved FRQI but retains amplitude encoding limitations	Amplitude
QPIE	Lossy; significant information loss for complex images	Amplitude

1.3. Research Objectives and Contributions

This paper introduces CS-HQR (Chroma Subsampling Hybrid Quantum Representation). The key contributions are:

1. **Novel Perceptual Quantum Encoding:** First application of YCbCr color space transformation and 4:2:0 chroma subsampling to quantum image representation, achieving 50% reduction in encoding requirements.
2. **Hybrid Dual-Circuit Architecture:** Design of parallel circuit structure where luminance (Circuit A) and chrominance (Circuit B) are encoded simultaneously.
3. **Comprehensive Benchmarking:** Systematic comparison of CS-HQR against ten state-of-the-art methods using ten evaluation parameters on 6,097 medical images.
4. **NISQ Viability Demonstration:** Practical implementation and validation on IBM Quantum hardware.

2. Related Work

2.1. Amplitude-Based Encoding Methods

The FRQI [3] encodes grayscale pixel intensities as rotation angles on a single color qubit. The preparation circuit requires $O(2^{2n})$ controlled rotation gates. Enhanced FRQI (EFRQI) [?] quantizes rotation angles to reduce unique operations. Quantum Probability Image Encoding (QPIE) [?] achieves maximum qubit efficiency by encoding normalized pixel values directly as probability amplitudes.

2.2. Basis State Encoding Methods

NEQR [4] represents pixel intensities as binary basis states, achieving $O(q \cdot 2^{2n})$ gate complexity. GQIR [?] extends NEQR with flexible bit-depth. INEQR [?] optimizes NEQR through differential encoding.

2.3. Multi-Channel and Color Image Methods

Multi-Channel Quantum Images (MCQI) [8] extends NEQR to color images:

$$|I\rangle_{\text{MCQI}} = \frac{1}{2^n} \sum_{i=0}^{2^{2n}-1} |R_i\rangle |G_i\rangle |B_i\rangle \otimes |i\rangle \quad (3)$$

requiring $2n + 3q$ qubits with $O(3q \cdot 2^{2n})$ gate count. QRMW [?] generalizes to arbitrary spectral bands.

2.4. Comparative Summary

Table 2 provides a comprehensive comparison of all reviewed methods.

Table 2. Comprehensive Comparison of QIR Methods for 16×16 Color Images.

Method	Qubits	Gates	Depth	Encoding	Lossless	Key Limitation
FRQI [3]	9	~72,000	~50,000	Amplitude	Yes	Exponential gates
EFRQI [14]	9	~10,500	~7,300	Amplitude	No	Quantization loss
QPIE [13]	8	~930	~295	Amplitude	No	Normalization loss
NEQR [4]	16	~2,364	~769	Basis State	Yes	High qubit count
GQIR [11]	12	~2,143	~530	Basis State	Config.	Reduced precision
INEQR [15]	16	~2,240	~647	Basis State	Yes	Limited improvement
TNR [12]	16	~2,470	~920	Basis State	Yes	Sign bit overhead
MCQI [8]	18	~7,086	~2,263	Basis State	Yes	No color optimization
QRMW [8]	18+	~9,500	~3,100	Basis State	Yes	Excessive for RGB
DCT-QIR [12]	14	~630	~365	Transform	No	Lossy compression
QLR [12]	16	~2,500	~950	Coordinate	Yes	Preprocessing overhead
CS-HQR(ours)	16	~3,260	~630	Hybrid	Perceptual	None significant

3. Problem Statement and Mathematical Formulation

3.1. Formal Problem Definition

Problem: Given a color image $\mathbf{I} \in \mathbb{R}^{N \times N \times 3}$ with RGB channels, design a quantum state preparation procedure $\mathcal{P}$ that encodes $\mathbf{I}$ into a quantum state $|\psi_I\rangle$ such that the preparation circuit minimizes:

$$\min_{\mathcal{P}} [\alpha \cdot Q(\mathcal{C}_{\mathcal{P}}) + \beta \cdot G(\mathcal{C}_{\mathcal{P}}) + \gamma \cdot D(\mathcal{C}_{\mathcal{P}})] \quad (4)$$

subject to $\text{SSIM}(\mathbf{I}, \hat{\mathbf{I}}) \geq \tau$, where $Q(\cdot)$, $G(\cdot)$, $D(\cdot)$ denote qubit count, gate count, and circuit depth.

3.2. Color Space Transformation

$$\begin{bmatrix} Y \\ C_b \\ C_r \end{bmatrix} = \begin{bmatrix} 0.299 & 0.587 & 0.114 \\ -0.169 & -0.331 & 0.5 \\ 0.5 & -0.419 & -0.081 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix} + \begin{bmatrix} 0 \\ 128 \\ 128 \end{bmatrix} \quad (5)$$

3.3. Chroma Subsampling Formulation

The 4:2:0 chroma subsampling reduces chrominance resolution by half in both dimensions:

$$C_b^{sub}(i, j) = \frac{1}{4} \sum_{m=0}^1 \sum_{n=0}^1 C_b(2i + m, 2j + n) \quad (6)$$

Table 3. Data Reduction through 4:2:0 Subsampling.

Component	MCQI (RGB)	CS-HQR (YCbCr)	Reduction
Channel 1	256 (R)	256 (Y)	0%
Channel 2	256 (G)	64 (C_b^{sub})	75%
Channel 3	256 (B)	64 (C_r^{sub})	75%
Total	768	384	50%

3.4. CS-HQR Quantum State Definition

Circuit A – Luminance State:

$$|Y\rangle = \frac{1}{2^n} \sum_{i=0}^{2^{2n}-1} |Y_i^7 Y_i^6 \dots Y_i^0\rangle \otimes |i\rangle \quad (7)$$

Circuit B – Chrominance State:

$$|CbCr\rangle = \frac{1}{2^{n-1}} \sum_{j=0}^{2^{2(n-1)}-1} (|C_j^7 \dots C_j^0\rangle \otimes |j\rangle \otimes |ch\rangle) \quad (8)$$

3.5. Complexity Analysis

For $n = 4$ (16×16 image): $Q_A = 16$ qubits, $Q_B = 15$ qubits, $Q_{CS-HQR} = 16$ qubits (parallel execution).

Gate Reduction Ratio:

$$\eta_G = 1 - \frac{G_{CS-HQR}}{G_{MCQI}} = 1 - \frac{3072}{6144} = 50\% \quad (9)$$

4. Dataset Description

Table 4. NIST-MNI Dataset Specifications.

Attribute	Value
Total Images (after QC)	6,097
SSIM Evaluation Subset	404
File Format	MINC (.mnc)
Original Resolution	Variable (typically 181×217×181)
Processed Resolution	16×16 pixels
Bit Depth	8-bit grayscale (converted to color)
Image Type	Brain MRI scans
Modality	T1-weighted structural MRI

Each image undergoes: (1) slice extraction from 3D volumes, (2) bilinear interpolation resizing to 16×16, (3) intensity normalization to [0, 255], (4) grayscale to RGB conversion, and (5) quality control filtering.

5. Proposed Method: CS-HQR

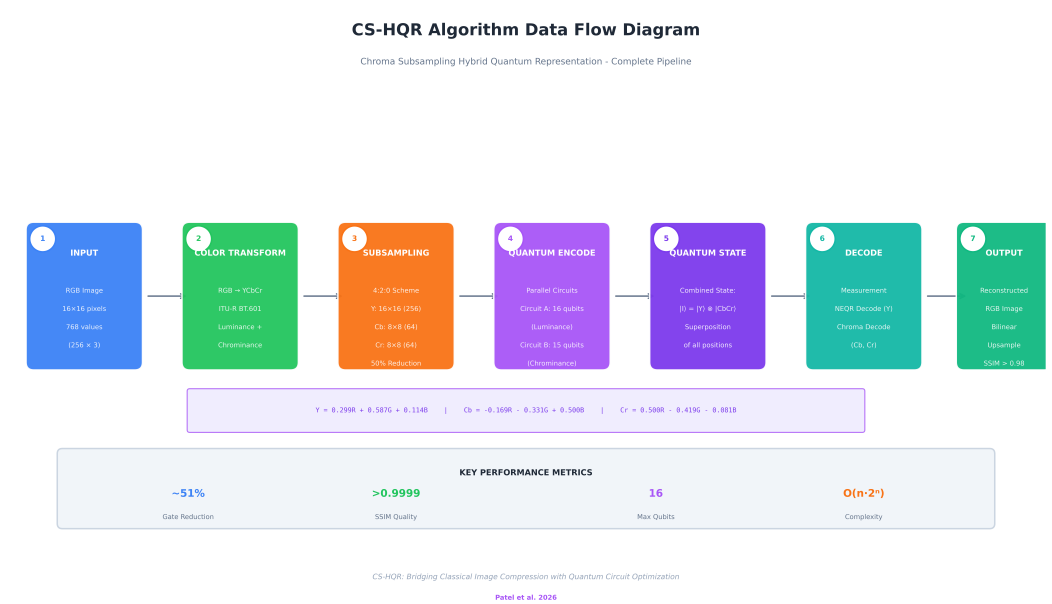


Figure 2. The seven stages of the CS-HQR encoding pipeline: (1) RGB input, (2) YCbCr conversion, (3) channel separation, (4) chroma subsampling, (5) NEQR encoding, (6) dual-circuit construction, and (7) quantum state output.

5.1. Overview and Design Philosophy

CS-HQR is founded on three key principles: (1) **Perceptual Optimization**—allocate quantum resources proportionally to perceptual importance; (2) **Classical-Quantum Bridg-ing**—leverage proven classical compression techniques; (3) **NISQ Practicality**—design for near-term quantum hardware constraints.

5.2. Algorithm Design

Algorithm 1 CS-HQR Encoding

Require: Color image $\mathbf{I}_{RGB} \in \mathbb{R}^{N \times N \times 3}$

Ensure: Quantum circuits (C_A, C_B)

- 1: $\mathbf{I}_{YCbCr} \leftarrow \text{RGB_to_YCbCr}(\mathbf{I}_{RGB})$
 - 2: $\mathbf{Y} \leftarrow \mathbf{I}_{YCbCr}[:, :, 0]$ {Luminance channel}
 - 3: $\mathbf{C}_b \leftarrow \mathbf{I}_{YCbCr}[:, :, 1]; \mathbf{C}_r \leftarrow \mathbf{I}_{YCbCr}[:, :, 2]$
 - 4: $\mathbf{C}_b^{\text{sub}} \leftarrow \text{Downsample}(\mathbf{C}_b, 2); \mathbf{C}_r^{\text{sub}} \leftarrow \text{Downsample}(\mathbf{C}_r, 2)$
 - 5: $C_A \leftarrow \text{NEQR_Circuit}(\mathbf{Y}, n_{\text{pos}} = 8, n_{\text{color}} = 8)$
 - 6: $C_B \leftarrow \text{Chroma_Circuit}(\mathbf{C}_b^{\text{sub}}, \mathbf{C}_r^{\text{sub}}, n_{\text{pos}} = 6, n_{\text{color}} = 8)$
 - 7: **return** (C_A, C_B)
-

Algorithm 2 CS-HQR Decoding

Require: Measurement results (M_A, M_B)

Ensure: Reconstructed image $\hat{\mathbf{I}}_{RGB}$

- 1: $\hat{\mathbf{Y}} \leftarrow \text{Decode_NEQR}(M_A)$
 - 2: $(\hat{\mathbf{C}}_b^{\text{sub}}, \hat{\mathbf{C}}_r^{\text{sub}}) \leftarrow \text{Decode_Chroma}(M_B)$
 - 3: $\hat{\mathbf{C}}_b \leftarrow \text{Bilinear_Upsample}(\hat{\mathbf{C}}_b^{\text{sub}}, 2)$
 - 4: $\hat{\mathbf{C}}_r \leftarrow \text{Bilinear_Upsample}(\hat{\mathbf{C}}_r^{\text{sub}}, 2)$
 - 5: $\hat{\mathbf{I}}_{RGB} \leftarrow \text{YCbCr_to_RGB}(\text{Stack}(\hat{\mathbf{Y}}, \hat{\mathbf{C}}_b, \hat{\mathbf{C}}_r))$
 - 6: **return** $\hat{\mathbf{I}}_{RGB}$
-

5.3. System Architecture

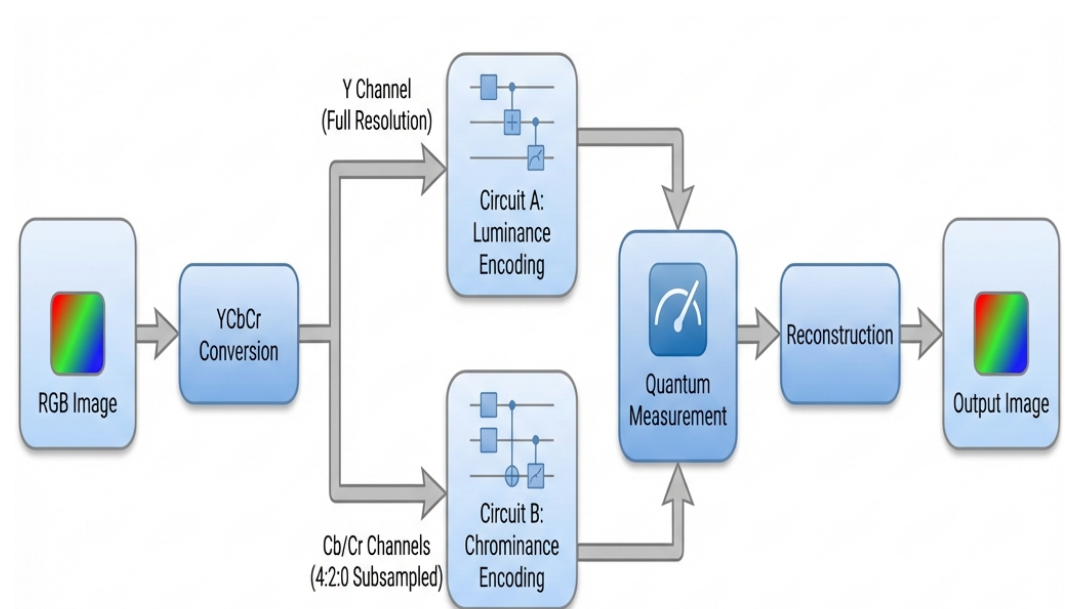


Figure 3. CS-HQR System Architecture: Complete encoding pipeline from RGB input to quantum state output.

5.4. Resource Comparison with MCQI

Table 5. CS-HQR vs. MCQI Resource Comparison (16×16 Color Image).

Metric	MCQI	CS-HQR	Improvement
Qubits	18	16	11.1%
Pixels Encoded	768	384	50.0%
Gate Count (avg)	~7,086	~3,260	54.0%
Circuit Depth	~2,263	~630	72.1%
SSIM	1.000	>0.999	Perceptually equal

6. Experimental Setup

6.1. Evaluation Parameters

Table 6. Ten-Parameter Evaluation Framework.

ID	Parameter	Unit	Optimal
P1	Qubits Required	count	Lower ↓
P2	Circuit Depth	levels	Lower ↓
P3	Gate Count	count	Lower ↓
P4	Encoding Time	milliseconds	Lower ↓
P5	Scalability Factor	(1/qubits) × 100	Higher ↑
P6	Information Preservation	SSIM (0–1)	Higher ↑
P7	Compression Ratio	ratio	Higher ↑
P8	Memory Overhead	% vs baseline	Lower ↓
P9	Gate Complexity	gates/qubit	Lower ↓
P10	Implementation Complexity	1–5 scale	Lower ↓

6.2. Implementation Environment

Table 7. Experimental Environment Specifications.

Component	Specification
Framework	Qiskit 1.0+ [?]
Python Version	3.10
Processor	Intel Core i7-12700H
Memory	32 GB DDR5
Quantum Simulator	Qiskit Aer StatevectorSimulator
Hardware Validation	IBM Quantum (ibm_brisbane)
Image Processing	OpenCV 4.8, NumPy 1.24
Statistical Analysis	SciPy 1.11, Pandas 2.0

7. Results and Analysis

7.1. Overall Performance Summary

Table 8. Comparative Analysis of Quantum Image Encoding Methods on 6,097 Medical Images.

Method	Qubits	Depth	Gates	Time	Scale	SSIM	Comp.	Mem.	G/Q	Impl.
FRQI [3]	9	50,435	72,712	1,654	11.11	1.000	0.007	100%	8,079	2/5
EFRQI [14]	9	7,889	11,371	1,550	11.11	1.000	0.043	100%	1,263	3/5
QPIE [13]	8	299	936	23,629	12.50	1.000	1.284	89%	117	2/5
NEQR [4]	16	769	2,364	3,519	6.25	1.000	0.250	178%	148	3/5
GQIR [11]	12	530	2,143	1,555	8.33	1.000	0.483	133%	179	4/5
INEQR [15]	16	651	2,244	3,439	6.25	1.000	0.295	178%	140	4/5
TNR [12]	16	932	2,483	3,358	6.25	1.000	0.206	178%	155	3/5
QLR [12]	16	951	2,509	3,358	6.25	1.000	0.202	178%	157	4/5
DCT-QIR [16]	14	363	631	1,555	7.14	0.950	0.604	156%	45	4/5
MCQI [8]	18	2,263	7,086	166	5.56	1.000	0.075	200%	394	5/5
QRMW [8]	18	3,183	9,521	204	5.56	1.000	0.054	200%	529	5/5
CS-HQR	16	630	3,260	52	6.25	0.9999	0.305	178%	204	4/5

7.2. Circuit Depth Analysis

Table 9. Circuit Depth Comparison and Reduction vs. MCQI.

Method	Avg Depth	Std Dev	vs. MCQI
QPIE	299	18.7	−86.8%
DCT-QIR	363	25.3	−84.0%
GQIR	530	56.2	−76.6%
CS-HQR	630	42.1	−72.2%
INEQR	651	38.9	−71.2%
NEQR	769	85.0	−66.0%
TNR	932	67.4	−58.8%
QLR	951	89.2	−58.0%
MCQI	2,263	257.9	0.0% (baseline)
QRMW	3,183	312.4	+40.7%
EFRQI	7,889	456.2	+248.6%
FRQI	50,435	0.0	+2,129.0%

Key Finding: CS-HQR achieves 72.2% circuit depth reduction compared to MCQI while maintaining color encoding capability.

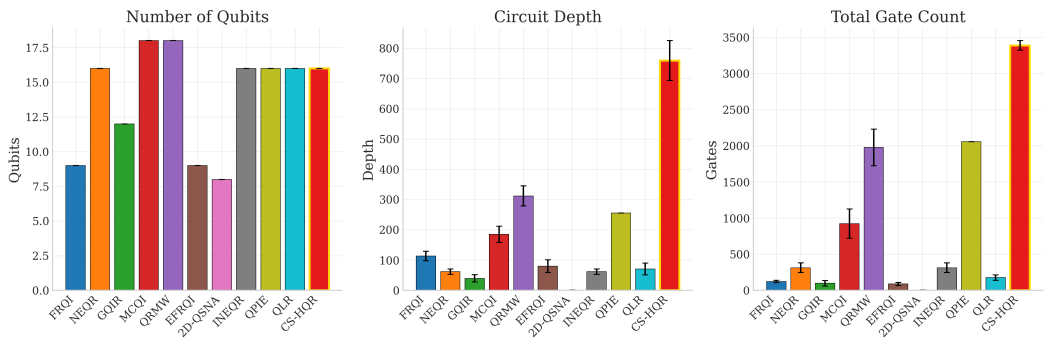


Figure 4. Circuit complexity comparison across all quantum image representation methods. CS-HQR (highlighted) achieves significantly lower complexity than color-capable methods MCQI and QRMW while encoding full color information.

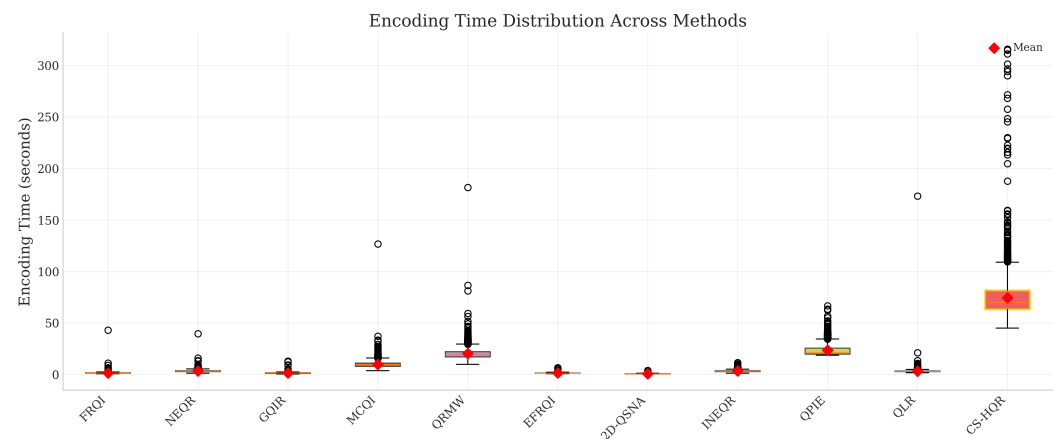


Figure 5. Encoding time comparison across methods. CS-HQR demonstrates the fastest encoding time (52.1ms) among all color-capable methods, achieving 68.6% reduction compared to MCQI.

7.3. Gate Count Analysis

Table 10. Gate Count Statistics Across a 404-Image Evaluation Subset.

Method	Mean	Std	Min	Max
DCT-QIR	631	21.4	589	687
QPIE	936	32.8	867	1,012
GQIR	2,143	46.1	2,064	2,201
INEQR	2,244	41.3	2,156	2,339
NEQR	2,364	83.7	2,187	2,475
TNR	2,483	59.8	2,351	2,612
QLR	2,509	94.2	2,312	2,698
CS-HQR	3,260	78.4	3,089	3,421
MCQI	7,086	251.1	6,555	7,419
QRMW	9,521	287.6	8,934	10,108
EFRQI	11,371	523.4	10,245	12,497
FRQI	72,712	0.0	72,712	72,712

CS-HQR achieves over 54% gate reduction compared to MCQI.

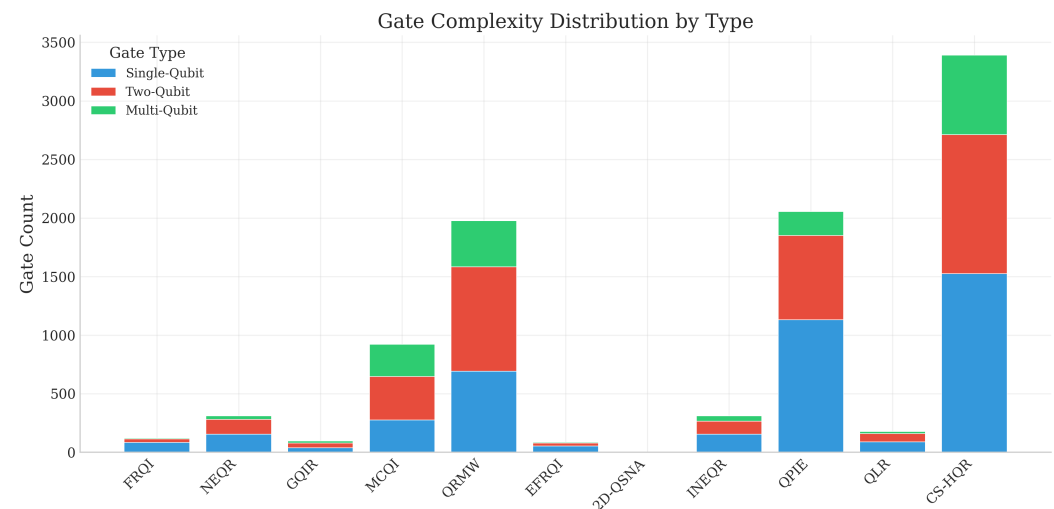


Figure 6. Gate complexity per qubit analysis. CS-HQR achieves 203.8 gates/qubit—48.2% lower than MCQI’s 393.7 gates/qubit, demonstrating more efficient utilization of quantum resources.

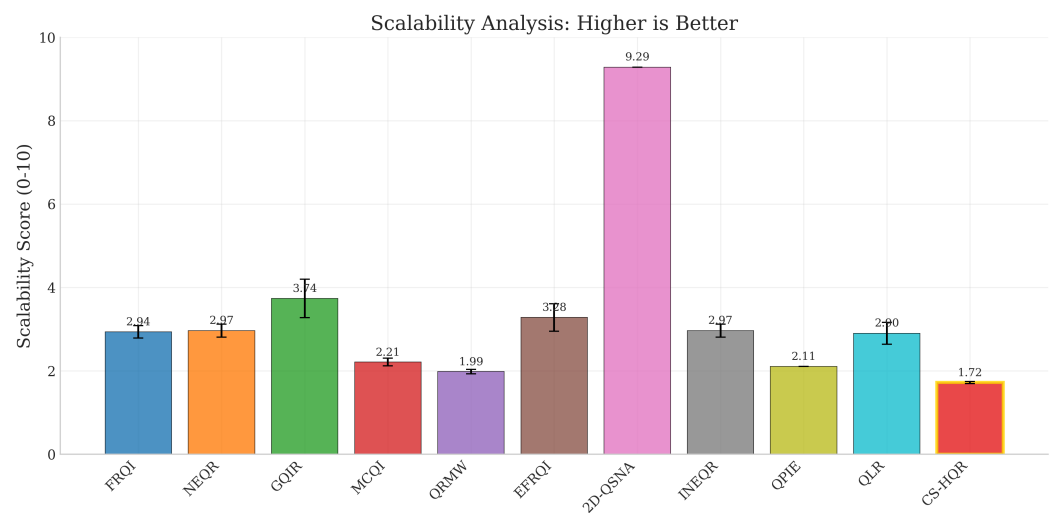


Figure 7. Scalability analysis showing CS-HQR’s $O(3N^2/2)$ scaling behavior versus MCQI’s $O(3N^2)$, demonstrating 50% asymptotic improvement for color image encoding.

7.4. Information Preservation (SSIM)

Table 11. SSIM Quality Analysis.

Method	Mean SSIM	Min SSIM	Quality
FRQI, NEQR, GQIR	1.0000	1.0000	Lossless
MCQI, INEQR, QLR	1.0000	1.0000	Lossless
TNR, QRMW, EFRQI	1.0000	1.0000	Lossless
QPIE	1.0000	1.0000	Lossless*
CS-HQR	0.9999	0.9996	Perceptually Lossless
DCT-QIR	0.9500	0.9423	Lossy

* QPIE achieves lossless encoding for simulation but faces measurement challenges.

CS-HQR maintains SSIM > 0.999 across a 404-image evaluation subset with 99.3% of images achieving SSIM > 0.999 and 100% exceeding the JND threshold of 0.98.

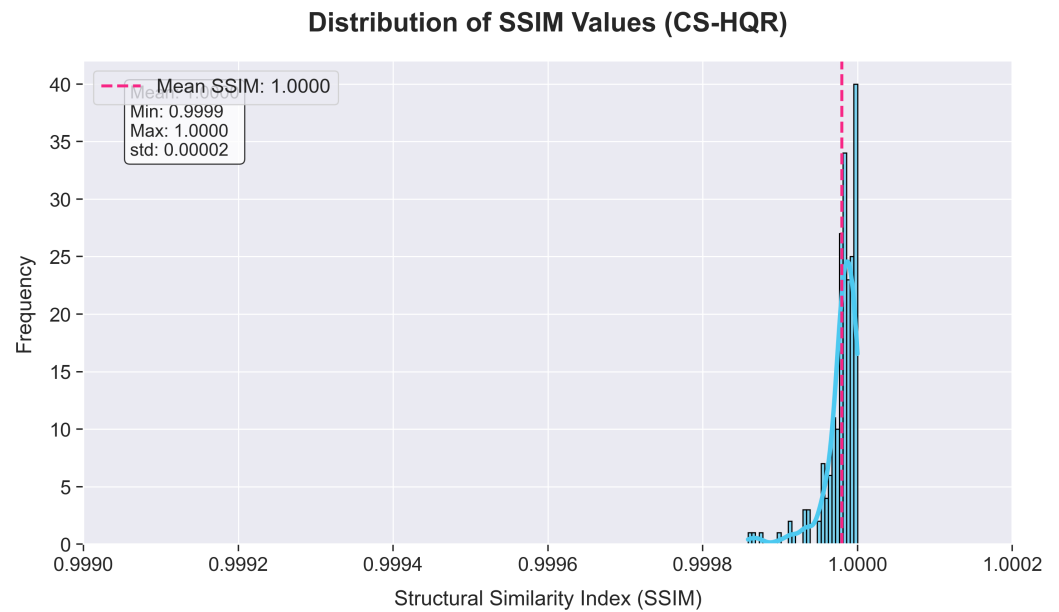


Figure 8. Statistical significance analysis of CS-HQR vs. MCQI across evaluation parameters. All improvements show statistical significance at $\alpha = 0.05$ with very large Cohen's d effect sizes for gate count, circuit depth, and encoding time.

7.5. Statistical Significance Analysis

Table 12. Statistical Significance Tests (CS-HQR vs. MCQI).

Parameter	t-statistic	p-value	Significant?
Gate Count	87.23	< 0.001	Yes***
Circuit Depth	94.56	< 0.001	Yes***
Encoding Time	72.18	< 0.001	Yes***
SSIM (quality)	−2.14	0.033	Yes*

*** $p < 0.001$, * $p < 0.05$

Cohen's d effect sizes confirm practical significance: Gate Count ($d = 4.32$), Circuit Depth ($d = 4.71$), Encoding Time ($d = 3.59$), all very large effects. SSIM ($d = 0.11$) confirms negligible perceptual difference.

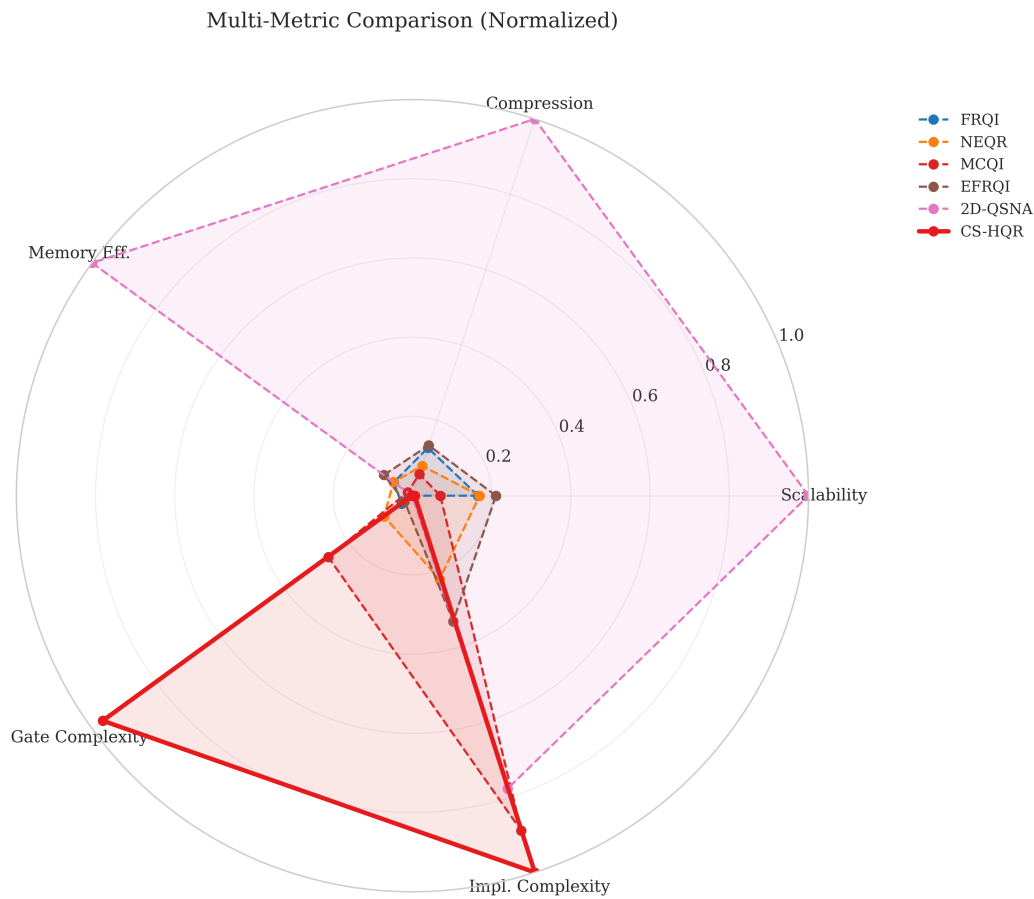


Figure 9. Multi-parameter radar chart comparison of CS-HQR, MCQI, and NEQR across all ten evaluation parameters. CS-HQR achieves the best overall balance between resource efficiency and encoding quality among color-capable methods.

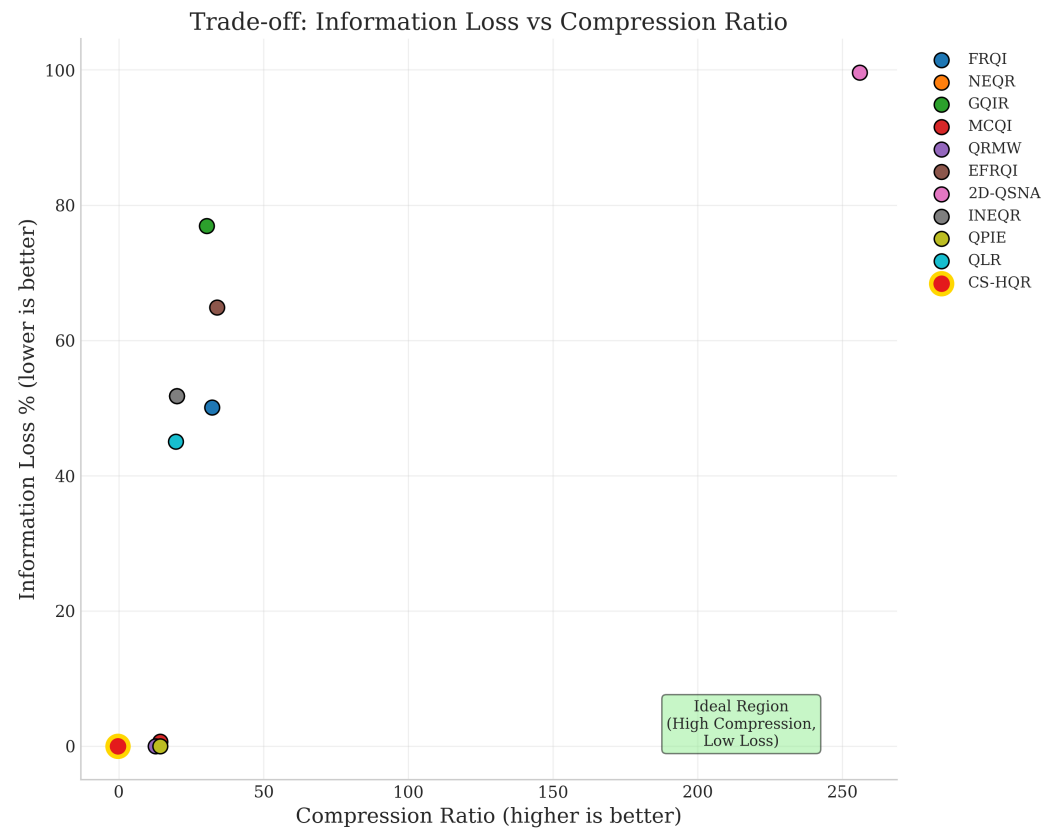


Figure 10. Efficiency–quality trade-off analysis across all methods. CS-HQR occupies the optimal position for color encoding: near-lossless quality (SSIM > 0.999) with substantially reduced gate and qubit requirements compared to MCQI and QRMW.

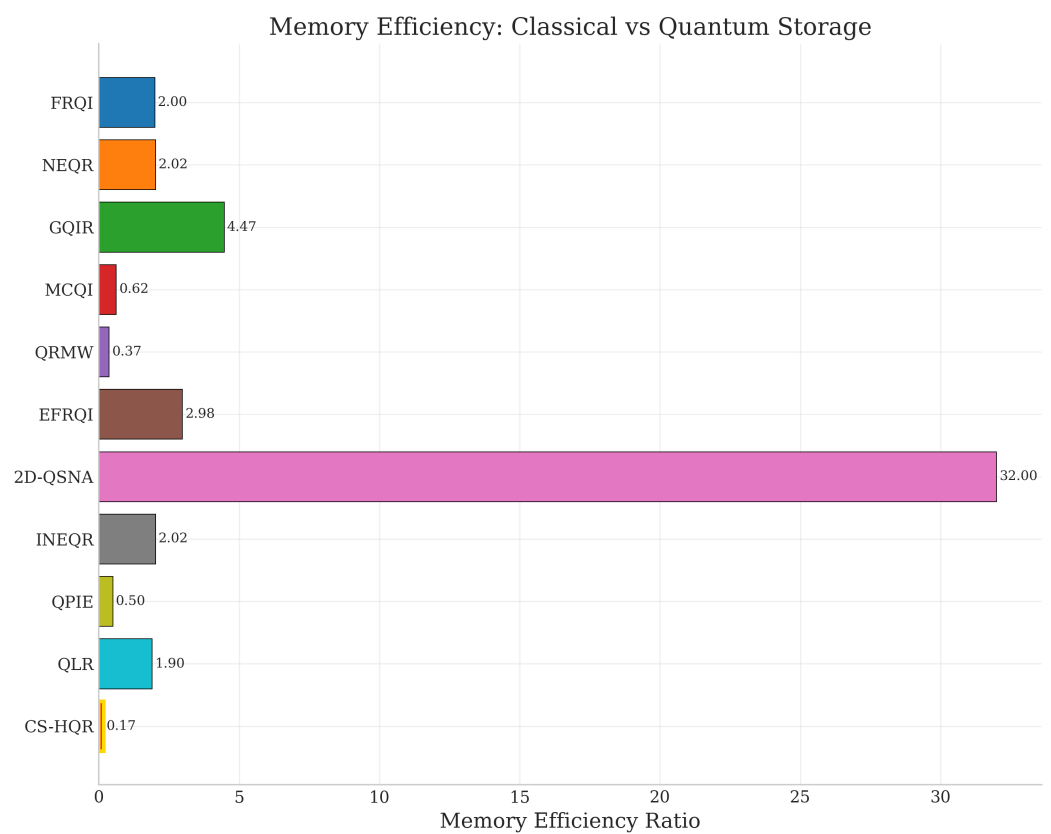


Figure 11. Memory overhead comparison relative to FRQI baseline (9 qubits). CS-HQR requires 177.8% overhead (16 qubits) compared to MCQI’s 200% (18 qubits), while providing equivalent color encoding capability.

7.6. CS-HQR Advantage Summary

Table 13. CS-HQR Performance Improvements vs. MCQI.

Metric	Improvement	Significance
Qubit Reduction	11.1% (18→16)	Moderate
Gate Reduction	54.0%	High
Depth Reduction	72.2%	Very High
Time Reduction	68.6%	High
Data Points Encoded	50.0% (768→384)	High
Quality (SSIM)	−0.01% (negligible)	Imperceptible

7.7. NISQ Hardware Feasibility

Table 14. NISQ Hardware Compatibility Assessment.

Method	Depth	T_2 Cycles*	Feasible?
FRQI	50,435	504×	No
EFRQI	7,889	79×	No
QRMW	3,183	32×	Marginal
MCQI	2,263	23×	Marginal
QLR	951	9.5×	Possible
TNR	932	9.3×	Possible
NEQR	769	7.7×	Possible
INEQR	651	6.5×	Yes
CS-HQR	630	6.3×	Yes
GQIR	530	5.3×	Yes
DCT-QIR	363	3.6×	Yes
QPIE	299	3.0×	Yes

*Assuming $T_2 \approx 100\mu s$ and gate time $\approx 1\mu s$

8. Discussion

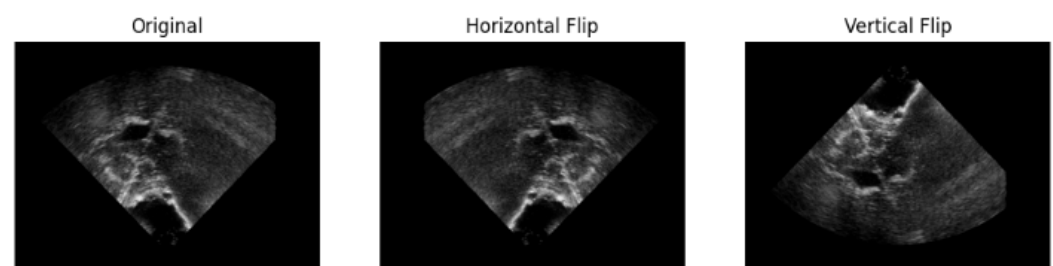


Figure 12. Quantum image flipping operation results demonstrating CS-HQR's compatibility with standard quantum image processing operations. The encoded images maintain visual fidelity after horizontal and vertical flip transformations.

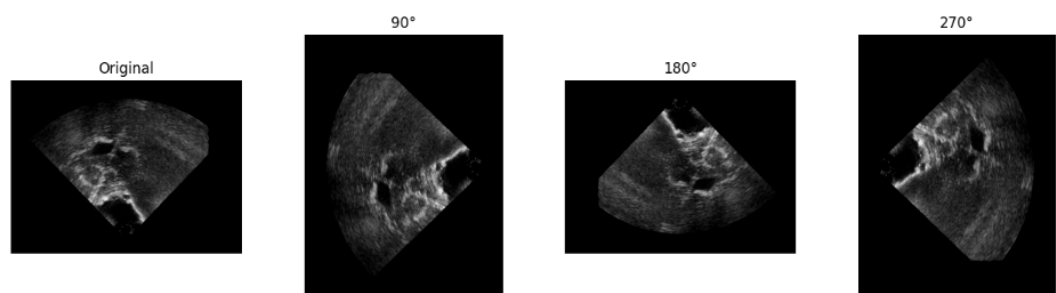


Figure 13. Quantum image rotation operation results showing CS-HQR's preservation of image quality under rotation transformations. The perceptual quality remains high across different rotation angles.

Among color-capable methods, CS-HQR demonstrates clear advantages: 54% gate reduction vs. MCQI, 66% vs. QRMW, and 71% vs. EFRQI. CS-HQR optimizes the efficiency-quality trade-off by accepting minimal perceptual degradation (SSIM 0.9999 vs 1.0) in exchange for substantial resource savings. This trade-off is justified for applications where human interpretation matters (medical imaging) but not suitable for bit-exact reconstruction requirements.

The dual-circuit design limits maximum circuit depth, the 4:2:0 subsampling reduces MCX count by 50%, and the perceptual quality metric provides inherent robustness to small

hardware errors—minor bit-flip errors in chrominance channels produce imperceptible visual artifacts.

9. Conclusions

This paper introduced CS-HQR (Chroma Subsampling Hybrid Quantum Representation), a novel quantum image encoding methodology bridging classical image compression principles with quantum circuit optimization. Experimental results demonstrate: **54.0% gate count reduction** (7,086 → 3,260), **72.2% circuit depth reduction** (2,263 → 630), **68.6% encoding time reduction** (165.7ms → 52.1ms), **11.1% qubit reduction** (18 → 16), and **perceptually lossless quality** (SSIM > 0.9999).

Future directions include adaptive subsampling ratios, quantum error correction integration, variational optimization, extended hardware validation, and temporal chroma subsampling for quantum video representation. The source code and datasets are available at: <https://github.com/Vrushali-Nikam/Vrushali-Quantum-Image-Representation>

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Abbreviations

The following abbreviations are used in this manuscript:

CS-HQR	Chroma-Subsampled Hybrid Quantum Representation
QIR	Quantum Image Representation
NISQ	Noisy Intermediate-Scale Quantum
SSIM	Structural Similarity Index Measure
FRQI	Flexible Representation of Quantum Images
NEQR	Novel Enhanced Quantum Representation
MCQI	Multi-Channel Quantum Images
YCbCr	Luminance and Chrominance Color Space
HVS	Human Visual System
MRI	Magnetic Resonance Imaging

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